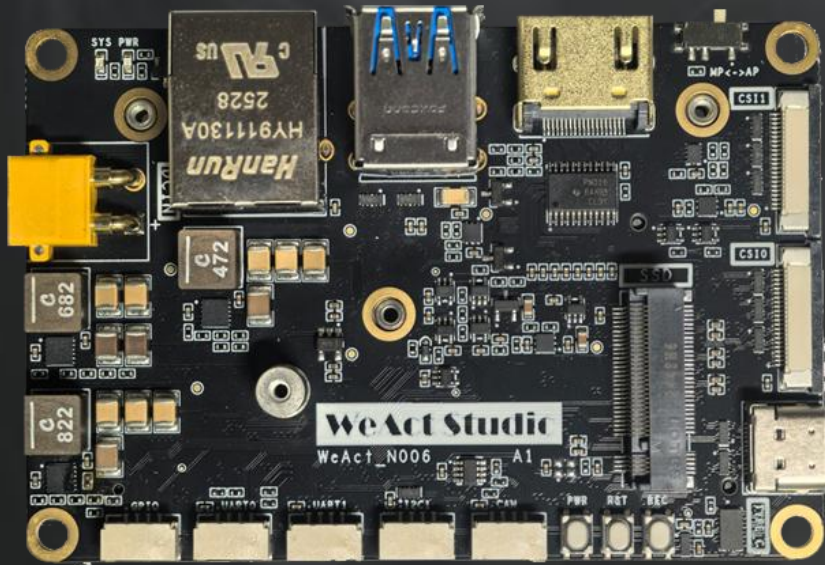




WeAct Studio

N006

CARRIER BOARD

User Guide

Catalog

Revision History	3
1. Disclaimer	4
2. Product Parameter.....	5
3. Application Scenario.....	7
4. Function Supported List	8
5. Hardware Interfaces Description.....	9
6. Hardware Parametes.....	10
7. Electrical	12
8. Mechanical Size	13

REVISION HISTORY

Draft Date	Revision	Description	Hardware
2026.02.17	V1.0	1. Initial	A1

1. DISCLAIMER

Welcome to the WeAct_N006 carrier board (hereinafter referred to as the "carrier board"). Before using the carrier board, please carefully read and fully understand all the terms of this disclaimer. Once you open or use the carrier board, it shall be deemed that you have voluntarily accepted and strictly adhered to all the agreements set forth in this statement.

The carrier board is manufactured after undergoing strict testing and debugging, complying with national electronic product quality standards and industry regulations. However, the company shall not bear any direct or indirect responsibility for product damage caused by failure to follow the operational procedures specified in this user manual, operational errors, disregard of product warning labels, or use under conditions not stipulated by the product (such as extreme temperatures, humidity, strong electromagnetic interference, etc.), as well as any resulting personal injuries or property losses.

The carrier board only supports two core boards: NVIDIA Jetson Orin NX 8G/16G (P3767-0000/P3767-0001). If damage to the core board occurs due to the use of other core boards, the company shall not bear any direct or indirect responsibility.

2. PRODUCT PARAMETER

- Support **NVIDIA Jetson OrinNX 8G/16G Core Board**; (*Note: Using other core boards may cause damage to the core board)
- The power on sequence of the full power supply is designed in strict accordance with NVIDIA recommendations, with discharge circuit;
- Supported DC XT30 (**16-28V**) [**Supported 6S Battery**];
- The power inlet is equipped with undervoltage, overvoltage, overcurrent and anti reverse connection protection, which makes it more safe to use;
- Support one Gigabit adaptive network port for network debugging, data communication, etc;
- Supports 2 USB 3.2 Gen 2x1 (10Gbps) ports for data transfer, sharing the USB 3.2 Gen 2x1 (10Gbps) bandwidth;
- Support 3 ports of USB2.0, of which 1 channel of OTG is used for system burning and data transmission, and the other 2 channels are host for data transmission;
- **Support Type C DRD, USB3.2 Gen2 bandwidth, and support HOST/Device switching;**
- Support 1 HDMI (2K) interface for screen display;
- Supports 1 CAN, 2 UART, 4 IO, and 1 I2C interfaces for more convenient data transmission;
- Supports 1 PCIe M.2 KEYM interface, PCIe 4.0 X 4 NVME protocol, 2242 size solid-state drive (SSD);
- Supports 1 PCIE M.2 KEYE interface, wireless WIFI network card;
- Support 2 port 4 Lane MIPI CSI Camera Interface, which can collect binocular camera data at the same time for binocular recognition;
- You can choose to power on with a button or automatically power on when powered on, to meet more application scenarios;

- All interfaces are equipped with ESD protection to prevent damage to the carrier plate caused by static electricity;
- The small carrier plate has compact structure, and the size is only 60mm * 90mm;
- Regularly update the device trees of different versions to be compatible with different Tegra kernel versions;

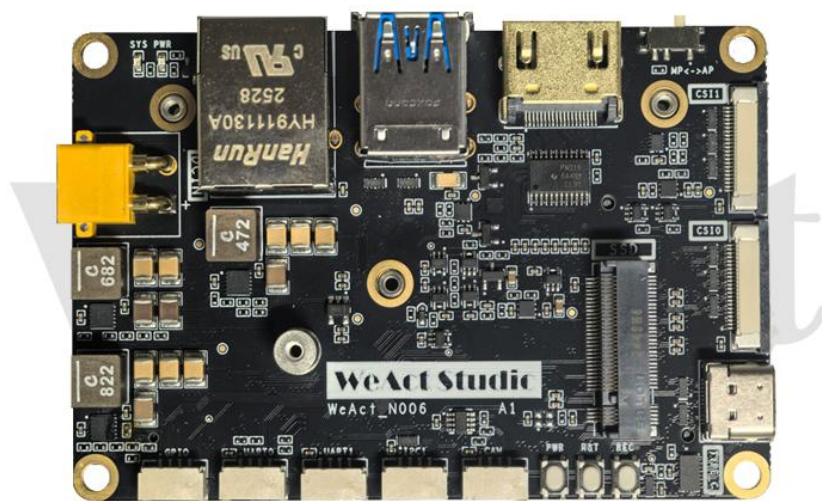


Figure.1 Top View

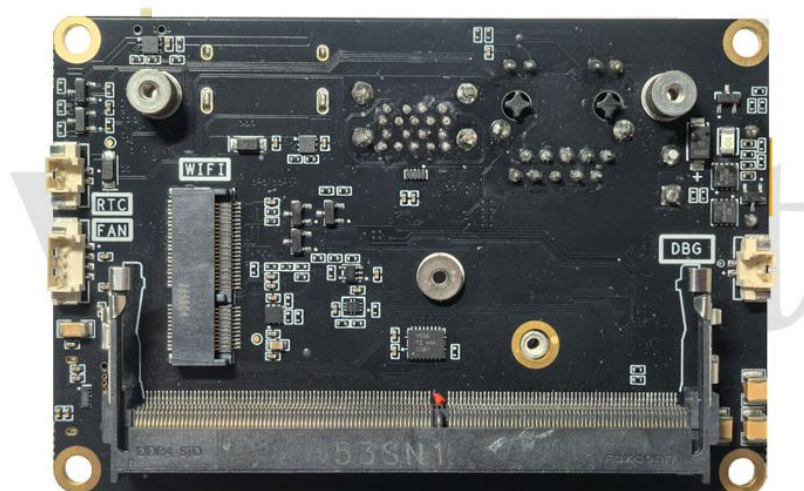


Figure. 2 Bottom View

3. APPLICATION SCENARIO

- √ Deep Learning
- √ Machine Vision
- √ Laboratory
- √ Robot Competition
- √ UAV
- √ Driverless
- √ AGV Navigation

4. FUNCTION SUPPORTED LIST

	OrinNX 8G	OrinNX 16G
GigaEthernet	√	√
2 * USB3.2 Gen2x1(10Gbps) [Type A]	√	√
HDMI	√	√
M.2 SSD (PCIe Gen4x4) [2242]	√	√
M.2 WIFI (PCIe Gen4x1) [2242]	√	√
2* CSI	√	√
USB2.0 OTG [Type C]	√	√
USB3.2 Gen2x1 DRD(10Gbps) [Type C]	√	√
4 * GPIO (3.3 V)	√	√
2 * UART (3.3V)	√	√
DEBUG_UART (1.8V)	√	√
I2C (3.3V)	√	√
CAN(3.3V) [H/L]	√	√
RTC(3.3V)	√	√
FAN Interfaces(5V)	√	√
Auto Power	√	√

▲ **Super Mode (40W)**

▲ **Power Protection:** Reverse Connection / UnderVoltage
OverVoltage / Shot Circuit Protection

▲ **Interface Protection:** ESD Protection

Figure 3. Function Support Lists

5. HARDWARE INTERFACES DESCRIPTION

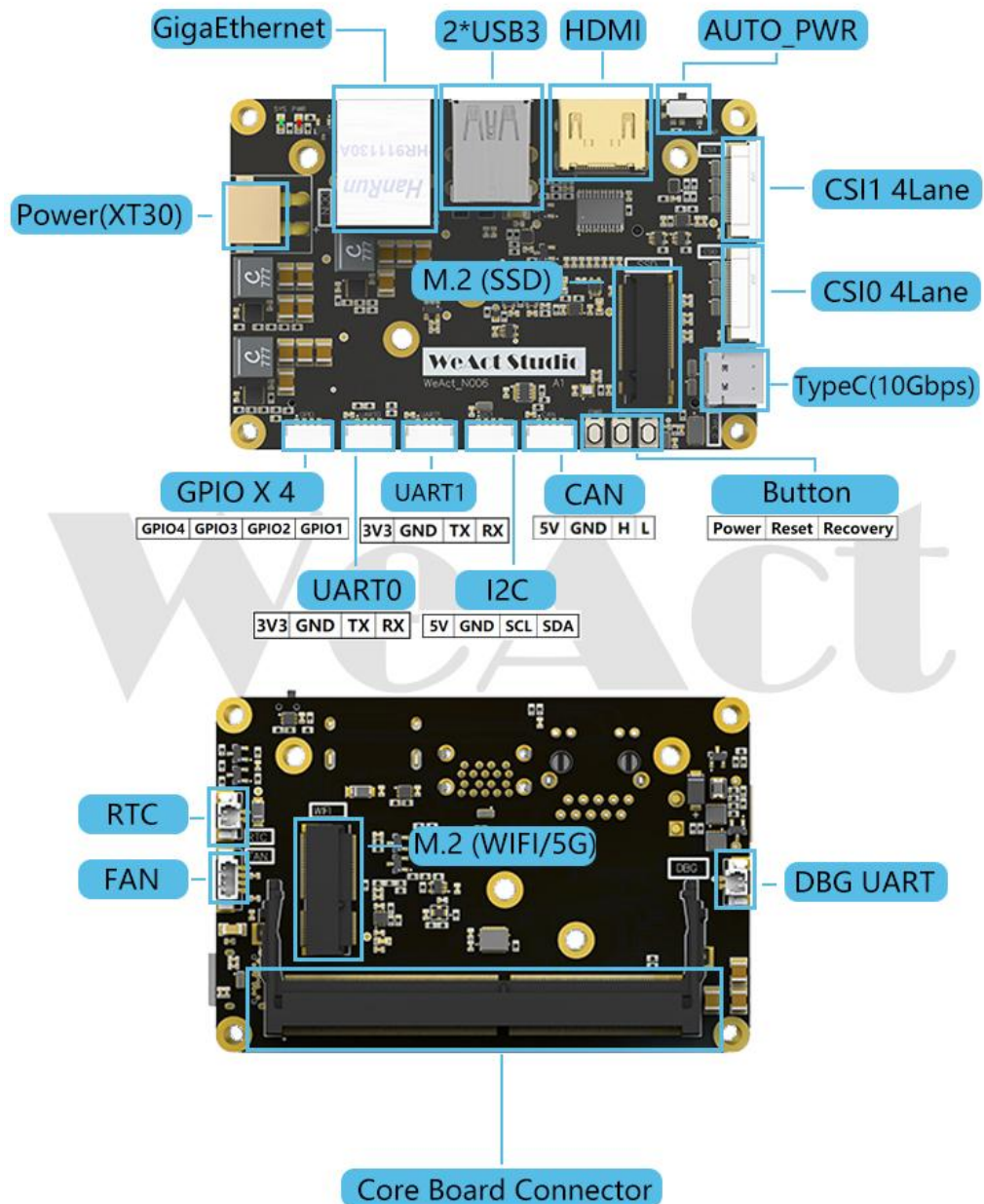


Figure. 4 Carrier Board Hardware Interfaces Descriptions

6. HARDWARE PARAMETES

KEY	1 x Power On Key 1 x Recovery Key 1 x Reset Key
LED	1 x Power LED (RED) 1 x System Status LED (Green)
USB	1 x USB3.1*2 Interface(10Gbps) 1 x USB2.0 OTG TypeC Interface 2 x USB2.0 HOST Interface(Same with USB3.2 Gen1 connector)
TypeC(USB)	1 x USB3.2 Gen2 TypeC(10Gbps)
CSI	2 x 4 Lane MIPI CSI Interface
PCIE	1 x PCIE M.2 KEYM(SSD) Interface 1 x PCIE M.2 KEYE(WIFI) Interface
HDMI	HDMI A
Ethernet	1 x 1000M RJ45
CAN	1 Port, 1 x 4P GH1.25 Connector
UART0/1	2 Port, 2 x 4P GH1.25 Connector
DBG_UART	1 Port, 1 x 2P GH1.25 Connector
GPIO	4 Port, 1 x 4P GH1.25 Connector
I2C	1 Port, 1 x 4P GH1.25 Connector
FAN	1 x 4P TX1.25 Connector
Auto Power On	1 x Side Switch
POWER	1 x XT30 Connector

RTC Power	1 x RTC Battery Interfaces
-----------	----------------------------

WeAct Studio

7. ELECTRICAL

Parameter	Minimum	Typical	Maximum
Temperature	0°C	/	40°C
Voltage	16V	24V	28V

8. MECHANICAL SIZE

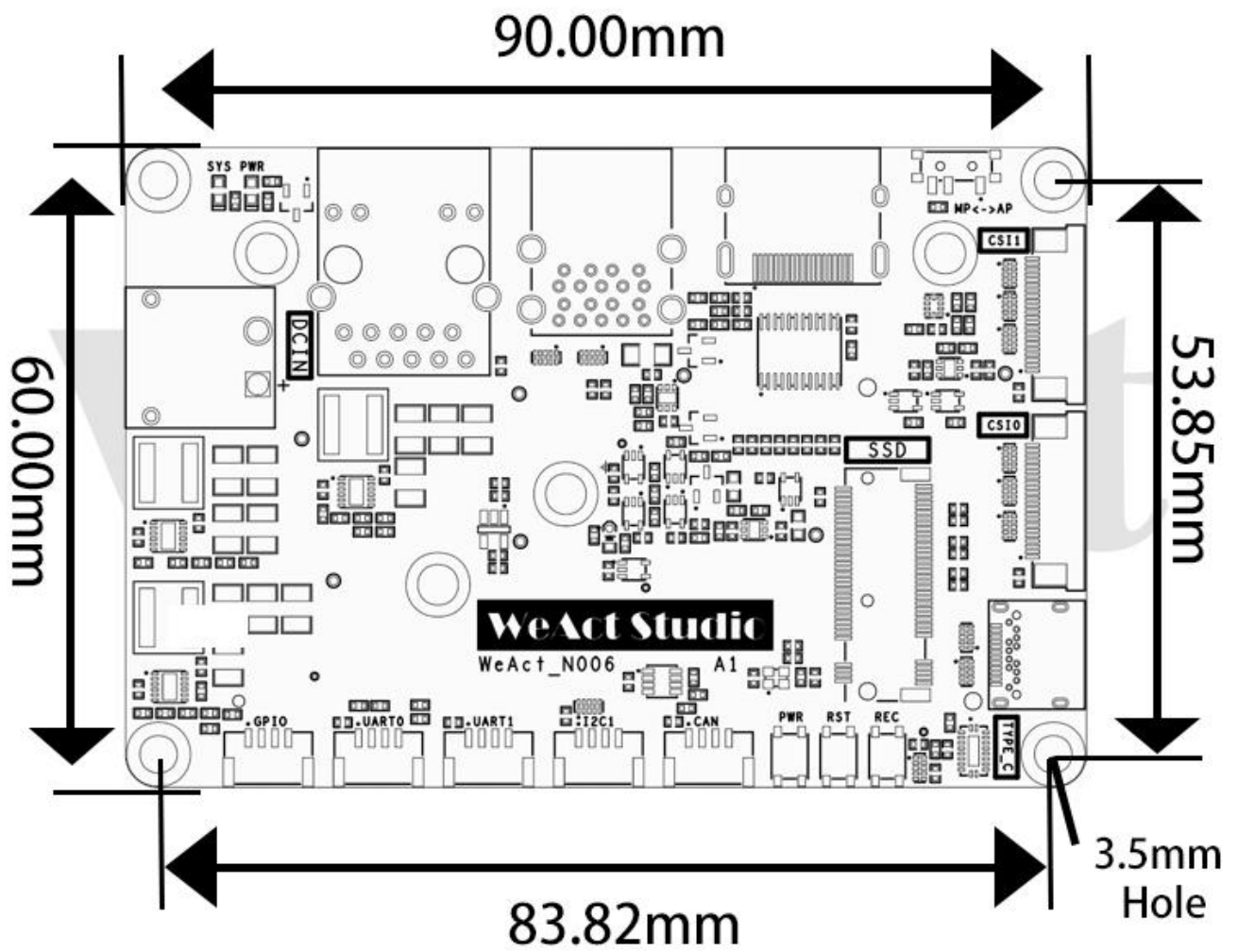


Figure 6. Carrier Baord Size